

LÊ NAM KHÁNH

AI / ML ENGINEERING | MULTIMODAL RETRIEVAL | RESEARCH

Information Technology student | University of Science, VNU-HCM (HCMUS)

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AI CHALLENGE 2026	HCMUS CODING	CODEFORCES	SOICT 2026
Finalist Bảng A (Top Teams)	Champion 2026	Expert Max 1796	Paper First Author

PROFILE

Information Technology student at HCMUS with a rigorous competitive programming background and practical expertise in multimodal video retrieval and AI systems. First & corresponding author of SOICT 2026 research on structured temporal event alignment (GEMTRA) and Finalist at the Ho Chi Minh City AI Challenge 2026. Skilled in architecting end-to-end retrieval pipelines, fast algorithmic reasoning under constraints, and full-stack engineering.

EDUCATION

University of Science, VNU-HCM (HCMUS) Information Technology Student	Current direction Multimodal video retrieval, temporal event reasoning, and foundation model systems.
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TECHNICAL SKILLS

Algorithms	Data structures, graph algorithms, dynamic programming, constraint optimization
Languages	Python, C++, TypeScript / JavaScript, Rust, SQL, Bash / PowerShell
AI / Retrieval	Visual embeddings (SigLIP2), FAISS, RF-DETR, OCR/ASR, VQA, GEMTRA alignment
Engineering	React 19 / Vite, Firebase / Firestore, Tauri, FastAPI, Git, testing, DRES API

ACHIEVEMENTS

AI Challenge HCMC 2026 Finalist (Bảng A) Team Lead, Reply 404	HCMUS Coding Challenge Champion (Rank 1) 2026
GDGoC AI Challenge 2026 Top 20 Outstanding Team RL Approach	KTC Ideathon 2026 Finalist (HrClaw) LIKELION & KOSME
SOICT 2026 Full Paper First & Corresponding Author (LNCS)	Codeforces Expert Max rating 1796
National Young Informatics 1st Prize (Central C2) & Finalist (2024)	Olympiad 30/4 & Prov. HSG Bronze Medal Prov. 2nd & 3rd Prizes

RELEVANT STRENGTHS

- Fast at translating mathematical formulations & constraints into efficient implementations. - Resilient operator with high debugging agility under live competition countdowns. - Rigorous engineering mindset: separating empirical evidence from benchmark claims.	- Full-stack capability: bridging ML/AI backend systems with fast, intuitive user experiences. - Strong team collaborator across research ideation, system benchmarking, and code review. - English: IELTS 6.5 GitHub: lenamkhanhh Codeforces: Average2k7
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SELECTED PROJECTS

Reply 404 — Video Retrieval System Team Lead & Core Architect Python, SigLIP2, GEMTRA DP, DRES API Led 4-member team to design and build an interactive video search platform for 1,487 videos & 533K keyframes with sub-7ms temporal DP alignment (GEMTRA) and evidence-linked VQA under AIC countdowns. SOICT 2026 Paper GitHub / HCMAIC-Retrieval	TripFlow Workbench React 19 TypeScript Vite Firebase Auth & Firestore Rules Built a collaborative travel planning workbench featuring 4 realtime workflows: timeline scheduling with priority/assignees, expense splitting, and team member synchronization. Live App / mxhuit26.vercel.app GitHub
AI Account Switcher Tauri Rust React Local API Gateway CLI Wrappers Desktop app managing multi-account rotation for AI coding assistants (Claude Code, Codex) with an OpenAI-compatible local proxy gateway and real-time quota tracking. GitHub / ai-switcher	RESEARCH-ORIENTED INTERESTS Multimodal Video Retrieval Temporal Event Reasoning Vision-Language Models AI Agents Focusing on verifiable retrieval algorithms, multimodal alignment under resource constraints, and reproducible benchmarks.